Orbit Odyssey (BB Edition) Manual

This manual is based on the original Orbit Odyssey manual, with modifications and clarifications for the workshops and competitions that will be held at Bloomberg.

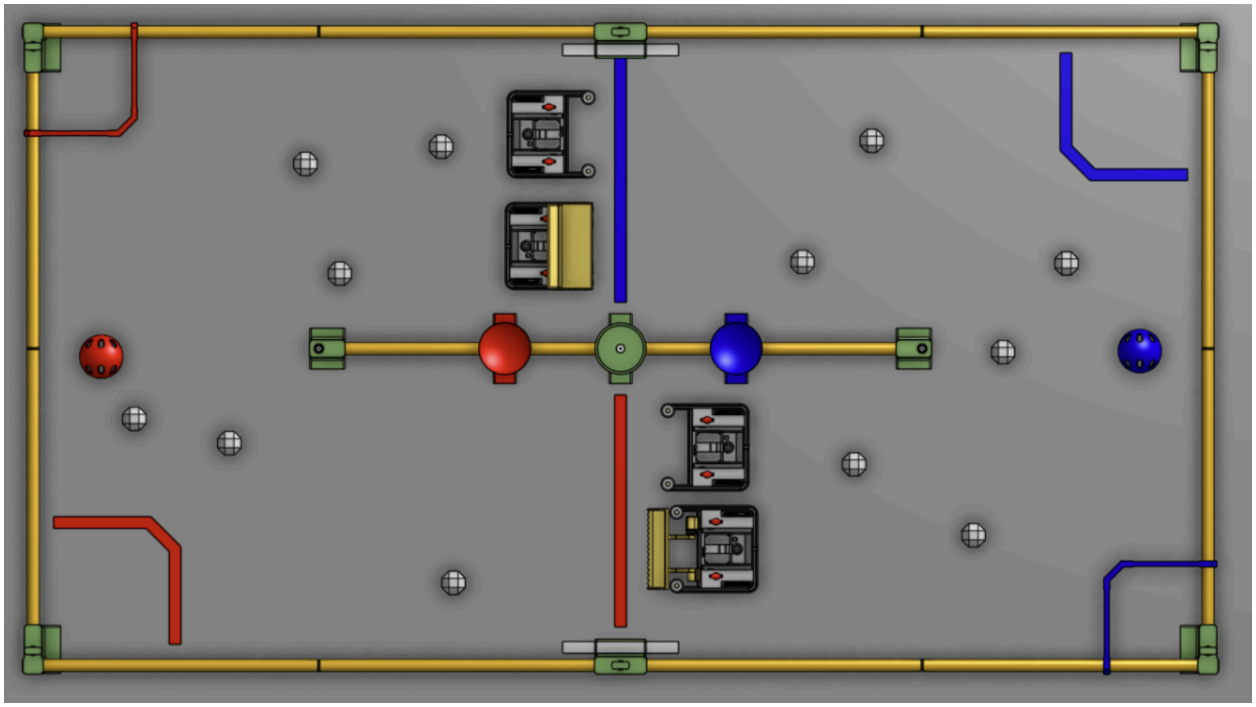
Last Updated: 10/20/2025

Quick Access Links

Game Animation	Original Orbit Odyssey Manual
https://bit.ly/oo-animation	https://bit.ly/orbit-odyssey-manual

1 Game Overview

In Orbit Odyssey, 2 ALLIANCES (an ALLIANCE is a cooperative of up to 2 teams) are tasked with cleaning up space RUBBLE and completing ORBITs. Teams score points by depositing RUBBLE in the LOW or HIGH RUBBLE ZONEs, completing ORBITs of the course, as well as parking in specific zones as the match concludes.



1. The challenge will begin with a 30-second AUTONOMOUS period (See starting positions above)
   1.1. Robots can score points by navigating to the LOW RUBBLE ZONE or HIGH RUBBLE ZONE, scoring preloaded RUBBLE. Each piece of RUBBLE is worth 2 points in the LOW RUBBLE ZONE, and 4 points in the HIGH RUBBLE ZONE.
   1.2. Robots that position themselves at the end of the AUTONOMOUS period (even partially) in the LOW RUBBLE ZONE score an additional 5 points.

2. After the 30-second AUTONOMOUS phase, teams will be given 60-seconds to transition to the TELEOPERATED phase. During this transition period, teams will reposition their robots on the field in the STARTING POSITIONS (See image above). Only robots may be repositioned during the transition period.

3. The 2-minute TELEOPERATED period will then begin.
   3.1. During the TELEOPERATED period, robots will race around the earth to gather data and collect space rubble. Robots may carry an AMPLIFIER with them to gather more data, earning bonus points.
      3.1.1. For each LAP a robot completes, it will earn 5 points

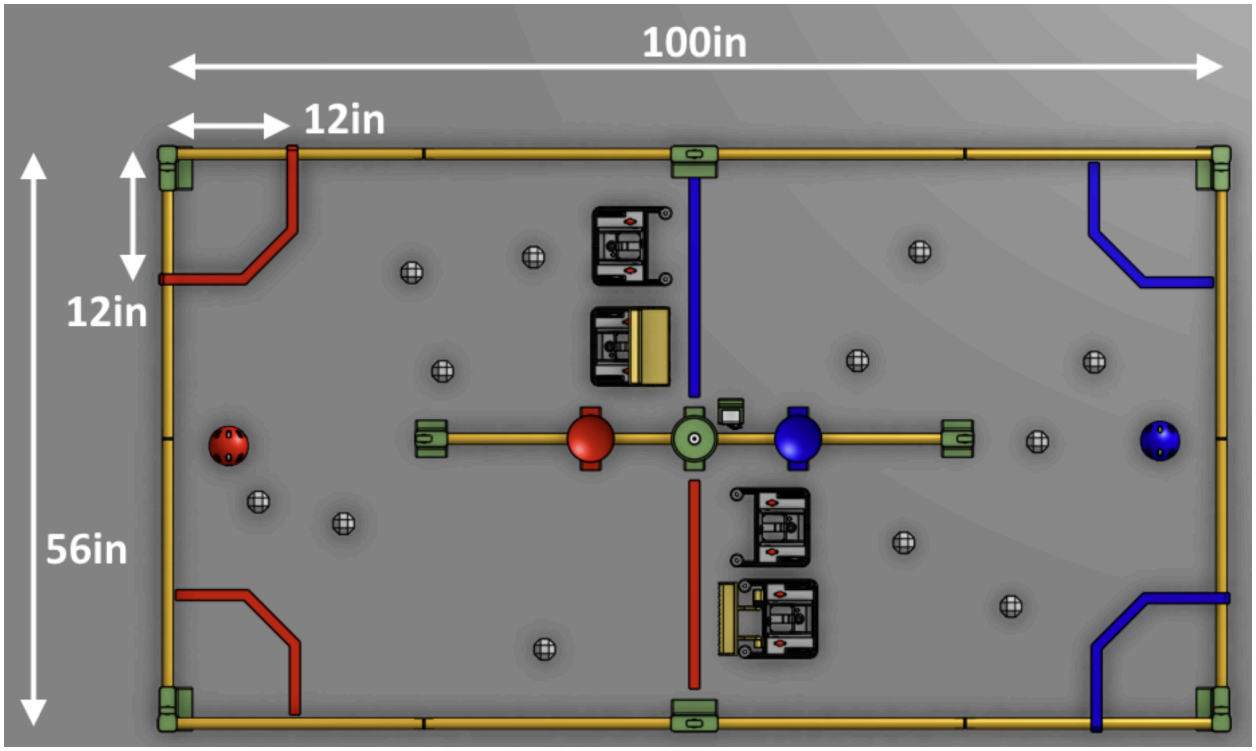
3.1.2. For each LAP a robot completes while holding an AMPLIFIER, it will earn 3 extra points, for a total of 8 points per AMPLIFIED lap.
3.1.3. Each piece of rubble scored in the LOW RUBBLE ZONE will score 1 point
3.1.4. Each piece of rubble scored in the HIGH RUBBLE ZONE will score 2 points.

4. The last 30-seconds is the ENDGAME
   4.1. An ALLIANCE will earn 10 points if its amplifier is returned to Earth (on the pedestal at the end of the match)
   4.2. Robots may also earn points for parking (even partially) in the LOW RUBBLE ZONE, worth 5 points.
   4.3. If both AMPLIFIERS are returned to Earth, there will be an additional 10 Coopertition points added to both teams scores.

Scoring Summary

		MATCH Points	
		AUTO	**TELEOP**
GAME PIECE	RUBBLE scored in LOW RUBBLE ZONE	2	1
	RUBBLE scored in HIGH RUBBLE ZONE	4	2
	AMPLIFIER returned to Earth		10
ORBIT	Robot ORBIT completion		5
	Robot ORBIT completion with AMPLIFIER		8
PARK	PARKING in the LOW RUBBLE ZONE of your ALLIANCE color	5	5
COOPERTITION	If both AMPLIFIERS returned to Earth		10

2 Field Overview



Scoring Locations

The Orbit Odyssey field contains several scoring locations.

1. RUBBLE ZONES
    1.1. Each ALLIANCE has a LOW and HIGH RUBBLE ZONE
        1.1.1. The HIGH RUBBLE ZONE is 12” x 12” and 2” tall
        1.1.2. The LOW RUBBLE ZONE is 12” x 12”

2. EARTH PEDESTAL
    2.1. Each EARTH PEDESTAL is 5” tall

Other Field Locations

1. CENTERLINE
    1.1. The CENTERLINE divides the field. There is a RED and BLUE section (See image)

Performing Orbits

1. An ORBIT is defined as a robot moving clockwise around the field 360 degrees
    1.1. ORBITs begin when a robot crosses the CENTERLINE while moving in a direction that would result in a clockwise motion around the field.
    1.2. ORBITs end when a robot crosses the CENTERLINE again, after having completed a clockwise motion around the field.
        1.2.1. It is possible to finish an ORBIT and begin a new one as part of the same movement

Game Pieces

1. RUBBLE
    1.1. Each RUBBLE is a white 3D print, with a nominal diameter of 2”

2. AMPLIFIER
    2.1. Each AMPLIFIER is an alliance specific colored Wiffle ball with a diameter of 3.5”

3 Robot Construction Rules

1. Each robot must follow these rules in order to pass inspection and compete.
    1.1. A robot in its starting configuration must fit inside a 10” x 10” x 10” cube
    1.2. A robot must weigh no more than 2.2 lbs

2. A robot may not contain materials or components that could cause damage to the playing field.

3. A robot should have a method of safely storing energy; for example, no launching rubber bands.

4. Robots must prominently display their ALLIANCE color and team number during matches
    4.1. Acceptable markers
        4.1.1. Blue or Red tape with team number clearly visible
        4.1.2. Clip on indicators (either team provided, or provided by event organizers)

5. Note: Additional rules may be necessary as the workshop season progresses. These will be published in a separate update document.

4 Tickets / Penalties

Breaching the following rules will result in a TICKET or a RED CARD

1. TICKET (2 points to the offended ALLIANCE)
    - 1.1. Pinning a robot for 5 or more seconds
    - 1.2. During AUTONOMOUS:
        - 1.2.1. Drive teams may not interact with the robot or operator controls
        - 1.2.2. After AUTONOMOUS starts, robots should remain on the side of the field with their corresponding RUBBLE ZONES
    - 1.3. Robots cannot enter the other alliance's RUBBLE ZONES
    - 1.4. A robot may not damage or functionally impair another robot or the playing field
    - 1.5. Robots may possess only 1 RUBBLE and/or AMPLIFIER at a time (1 ticket per additional RUBBLE/AMPLIFIER)
    - 1.6. Robots may not intentionally detach or leave parts on the field
    - 1.7. A robot may not launch, shoot or throw a game piece
    - 1.8. No one can enter the field during a match

2. RED CARD (Disqualification from the match)
    - 2.1. Excessive, ungracious, or unprofessional behavior will result in a red card
    - 2.2. Throwing RUBBLE or AMPLIFIERS at anyone is considered an automatic red card
    - 2.3. The team must send at least 1 team member to a match, whether the robot is there or not
    - 2.4. Repeated and egregious violations of TICKET-able offenses may result in a RED CARD

5 Match Play

Match Setup

1. During match setup, teams put their robots in their starting positions and preload RUBBLE if desired.
    1.1. Please make sure to avoid stepping on the field elements while putting robots onto the field.
2. Starting positions
    2.1. Robots begin the match with all their mechanisms within their starting configuration
    2.2. Robots must start with the chassis extending over the CENTERLINE and be facing their corresponding RUBBLE ZONES

AUTONOMOUS Period

1. Each match begins with a 30-second AUTONOMOUS period.
    1.1. A sound will play to signal that the AUTONOMOUS period has started.
    1.2. At the start of the AUTONOMOUS period, each driver should switch to the Autonomous mode in the Simulator Driver Station.
    1.3. After that, drivers may not touch their controls until the start of the TELEOP period.
    1.4. A sound will play to signal the END of AUTONOMOUS. At this point, drivers should switch to the Disabled mode in the Simulator Driver Station.

Transition from AUTONOMOUS to TELEOP

1. After AUTONOMOUS is completed, there will be an up-to-60-second transition period for referees and scorers to complete AUTONOMOUS scoring, and for teams to reset their robot positions in preparation for TELEOP.

TELEOP Period

1. After the transition period, a 2-minute TELEOP period will begin.
    1.1. A sound will play to signal that the AUTONOMOUS and Transition periods have ended, and the TELEOP period has begun.
    1.2. During TELEOP, drivers are free to touch their controls.
    1.3. Once there are 30 seconds left in the TELEOP period, a sound will play to signal that the ENDGAME phase has begun.
    1.4. A sound will play to signal that TELEOP has ended.
    1.5. Once TELEOP has ended, drivers must stop providing input to their robots, and switch to Disabled in the Simulation Driver Station.

6 Competition Setup

The competition will consist of randomly selected 2 vs 2 matches. The number of matches to be played will vary depending on how many teams are registered. The team with the highest average score after the final match is declared the event winner.

Judges will also be interviewing teams to ask about their approach to building a robot for the challenge. Awards may also be given out to teams based on programming, mechanical design, and other factors as a result of these interviews.

Change Log

Date	Change Description
10/20/2025	Initial Publication